Week3

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1 Policy Evaluation:

Aim: To evaluate a policy π

Solution (Pseudo Code):

- Start with all values as 0 i.e. $v_0(s) = 0 \forall s \in S$
- Update the value function for all states using the Bellman Equation:

$$v_{k+1}(s) = \Sigma_{a \in A} \pi(a|s)(\mathcal{R}_s^a + \Sigma_{s' \in S} P_{ss'} v_k(s'))$$

  In matrix form,

$$V^{k+1} = \mathcal{R}^\pi + \gamma V^k P^\pi$$

- Keep iterating until $|V^{k+1} - V^k| < \epsilon$
- The sequence $V^0, V^1, V^2, \cdots \to V^\pi$

2 Policy Iteration:

Aim: To improve the policy step by step and find the optimal policy.

Solution (Pseudo Code):

- Start with all values 0 and random Policy.
- Evaluate the value function at each state using policy evaluation algorithm mentioned above.
- Update the policy greedily i.e. at each state update the action to the action that maximizes the reward. If no state changes then the policy is stable.
- If policy is stable: stop, else: Go to step 1.

3 Value Iteration:

Aim: To improve the value function.

Solution (Pseudocode):

- Start with all values 0.
- Update the values as per the Bellman equation:

$$v(s) = max_a \Sigma_{s'} P^a_{ss'} [R^a_{ss'} + \gamma v(s')]$$

- Stop if $max(v(s)_{new} - v(s)_{old}) < \epsilon$. else Go to step 2.

4 Monte Carlo and Temporal differencing:

Although both of these methods are similar, model free and have the same aim, their answers often differ for finite set of sample. This is because of their different objectives. Monte Carlo minimizes the Mean Squared Error between the predicted value and the actual observed returns. While, Temporal Differencing builds an MDP that would explain the given samples. This can be demonstrated by a simple toy MDP as follows:

A 0 B 0 C 1
A 1
A 0 C 1
C 0
B 1
A 0 B 0
A 1 B 0
B 1 C 0

Monte Carlo:

$$v(A) = \frac{1+1+1+0+1}{5} = 4/5$$

$$v(B) = \frac{1+1+0+0+1}{5} = 3/5$$

$$v(C) = \frac{1+1+0+0}{4} = 2/4$$

Temporal Differencing (γ =1):

Since C is the terminal state. Its values is calculated exactly like Monte Carlo.

$$v(C) = 1/2$$

From B:

- Leads to C with reward 1 (once)

- Leads to C with reward 0 (twice)
- Leads to termination/nothing with reward 1 (once)
- Leads to termination/nothing with reward 0 (once)

From A:

- Leads to B with reward 0 (twice)
- Leads to termination with reward 1 (twice)
- Leads to C with reward 0 (once)

$$v(B) = (2/5)(\frac{1+0}{2}) + (3/5)(\frac{1+0+0}{3}) = 2/5$$

Similarly,

$$v(A) = (3/5)(\frac{0+0+1}{3}) + (1/5) * 0 + (1/5) * 1 = 2/5$$